MODULE 1 ASSIGNMENT

Networking Hardware, Home Wi-Fi Setup & Network Cables

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Course / Subject	Information Technology / Networking
Assignment Topic	Networking Devices and Cables

Purpose: To understand how common networking devices work together in a home network, identify common network cables, and compare the roles of routers and switches.

Prepared for learning and practical networking study

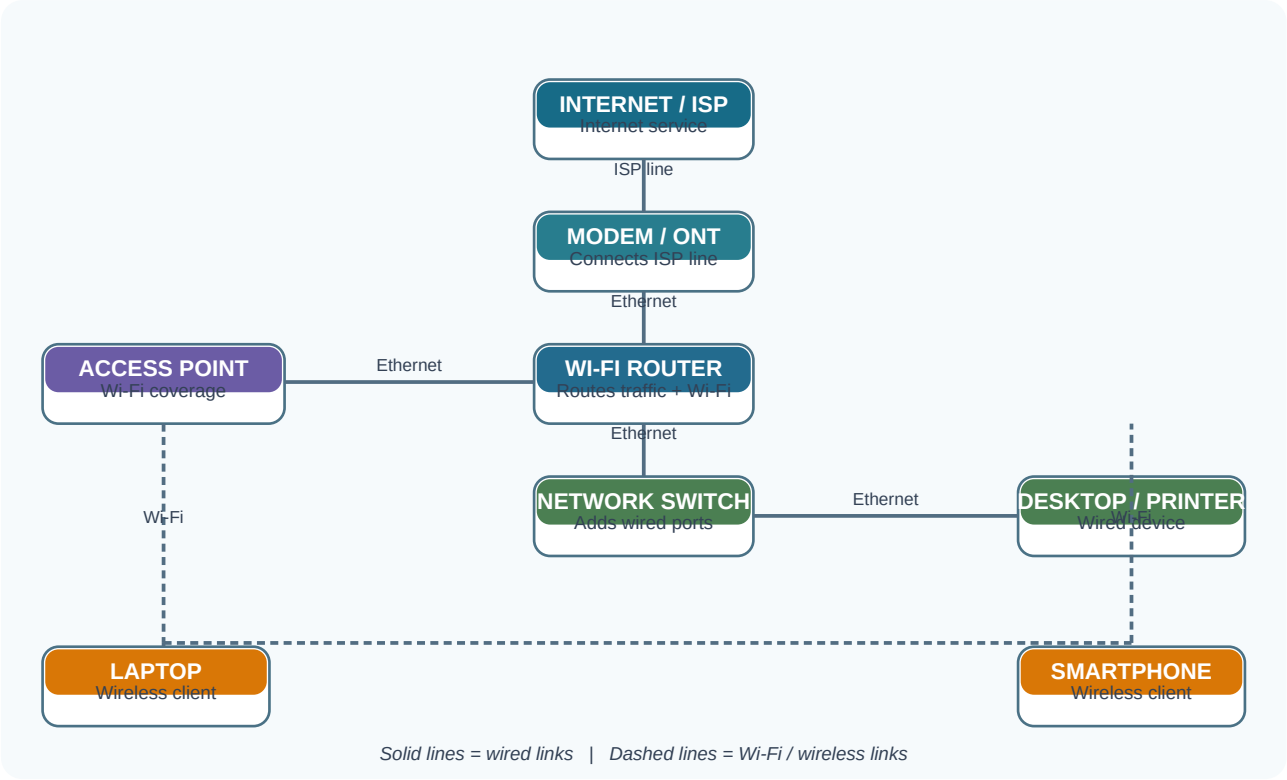
Introduction

A computer network is a group of devices connected so they can exchange information and share resources. A home network may include phones, laptops, smart TVs, printers, gaming consoles, and other devices. Some devices connect using Wi-Fi, while others use network cables.

The modem or optical network terminal (ONT) connects the home to the internet service provider (ISP). A router directs traffic between the home network and other networks. A switch provides more wired connections, and an access point allows Wi-Fi devices to connect to the wired network. In many homes, the router already contains a built-in switch and Wi-Fi access point, so separate devices are not always required.

1. Labeled Diagram of a Typical Home Wi-Fi Setup

The diagram below shows a typical arrangement. Actual homes may use a combined modem-router unit or an ONT for fiber internet. A separate switch and access point are useful when more wired ports or wider Wi-Fi coverage are needed.



How to read the diagram: The ISP connection reaches the modem/ONT. The router connects the home network to the internet and manages traffic. The switch expands the number of wired Ethernet ports. The access point provides Wi-Fi coverage, allowing a laptop and smartphone to connect wirelessly. A desktop computer or printer can connect to the switch by Ethernet cable.

2. Main Functions of Networking Hardware

Device	Main function	Real-life example
Modem / ONT	Connects the home connection to the ISP network. A modem converts signals for the service being used; an optical network terminal (ONT) terminates a fiber connection.	A cable modem or fiber ONT installed near the internet entry point in a house.
Router	Connects different networks and forwards IP packets toward their destinations. A home router commonly provides NAT, DHCP, basic firewall features, and often Wi-Fi.	The Wi-Fi router near a television or in a central room that shares internet access with family devices.
Network switch	Connects several wired devices on the same local area network (LAN). It learns device MAC addresses and forwards Ethernet frames to the appropriate port when possible.	A 5-port or 8-port switch in a home office connecting a desktop, printer, smart TV, and access point.
Access point (AP)	Provides a wireless connection for Wi-Fi devices and bridges them to the wired LAN. It can improve coverage where the router's Wi-Fi signal is weak.	A ceiling-mounted access point in a school, office, or a far room of a large house.
Network cable	Carries data between wired networking devices. Ethernet twisted-pair cable is common for short connections; coaxial and fiber cables are used for other access and backbone links.	An Ethernet cable from a router to a desktop PC, or a fiber cable carrying an ISP connection.

Important note: Modem, router, switch, and access point describe different networking functions. Consumer devices often combine several of these functions in one box. For example, an ISP-provided home gateway may include a modem, router, Ethernet switch, and Wi-Fi access point.

A simple way to remember them

- Modem/ONT: brings the ISP connection into the home.
- Router: connects networks and decides where IP traffic should go.
- Switch: connects wired devices within the local network.
- Access point: connects Wi-Fi devices to the local network.
- Network cable: physically carries data between compatible ports.

3. Three Common Types of Network Cables

Network cables carry signals between devices. The cable type depends on the service, distance, speed, environment, and equipment connectors. The following descriptions are suitable for a basic networking assignment. Source links are provided so you can view reference images and diagrams online.

A. Ethernet cable (twisted pair)

Usually contains pairs of copper wires twisted together and commonly uses an RJ-45-style connector. It connects PCs, routers, switches, access points, smart TVs, and other LAN devices.

Common use: wired connections in homes, computer labs, and offices.

Reference images / source: https://e115.engr.ncsu.edu/networking/network-media/

B. Coaxial cable

Has a central conductor, insulating layer, metallic shield, and outer jacket. The shielding helps reduce outside interference.

Common use: cable television and some cable-broadband internet connections.

Reference images / source: https://e115.engr.ncsu.edu/networking/network-media/

C. Fiber-optic cable

Carries data as pulses of light through very thin glass or plastic fibers. It supports high capacity and long-distance communication and is not affected by electromagnetic interference in the same way as copper cable.

Common use: fiber-to-the-home internet, ISP networks, data centers, and long-distance links.

Reference images / source: https://e115.engr.ncsu.edu/networking/network-media/

Image note: The source page includes illustrations or photographs of these cable types. If your instructor requires original photographs, take clear photos of cables you can access safely, or add images from a source that permits educational reuse and include its credit.

4. Router vs. Switch — Comparison Table

A router and a switch are both important networking devices, but they solve different problems. A router connects networks, while a switch connects devices within a local network. The comparison below describes common home and small-office equipment; advanced models may offer extra functions.

Comparison point	Router	Switch
Main use	Connects different networks, such as a home LAN and the internet.	Connects wired devices within the same LAN.
Typical location	Near the modem/ONT or the point where internet enters the home.	In a home office, computer lab, server room, or wiring cabinet.
How it handles data	Forwards IP packets using destination IP addresses and routing information.	Forwards Ethernet frames using MAC-address information and a MAC table.
Typical connections	WAN/Internet port plus LAN ports; many home routers also provide Wi-Fi.	Multiple Ethernet ports for computers, printers, servers, access points, and other wired devices.
Addressing / decisions	Uses IP networks and routing tables to choose where packets should go.	Learns which MAC addresses are reachable through its ports and forwards frames accordingly.
Internet sharing	Commonly provides the gateway function, NAT, DHCP, and basic firewall features in home networks.	A basic Layer 2 switch does not normally provide internet routing or NAT by itself.
Wireless support	Home wireless routers usually include Wi-Fi, but not every router has wireless built in.	A standard Ethernet switch does not provide Wi-Fi; an access point can connect to it.
Example	A family router shares one internet connection with phones, laptops, and a TV.	An 8-port switch connects a desktop, printer, NAS, TV, and access point by cable.

Easy memory trick: a router is like a road junction between different places (networks). A switch is like a building's internal connection panel that lets rooms (devices) communicate. This is a simplified analogy, but it helps remember their main roles.

5. Example: How Data Travels in the Home Network

Imagine a student opens a learning website on a smartphone connected to home Wi-Fi. The smartphone sends its request over Wi-Fi to the access point (or the Wi-Fi function built into the router). The router forwards the request toward the internet through the modem or ONT and the ISP. The website's response returns through the ISP connection and router, then reaches the phone over Wi-Fi.

Now imagine a desktop computer connected to the network by Ethernet cable. Its data travels through the switch and router as required. If the destination is another local device, a switch may forward the traffic locally without sending it to the internet. The exact path depends on whether the devices are connected to a combined gateway, separate access point, and/or separate switch.

6. Basic Network Safety and Good Practice

- Change the default router administrator password to a strong, unique password.
- Use WPA2-AES or WPA3 Wi-Fi security where supported, and set a strong Wi-Fi password.
- Keep router and access point firmware updated.
- Place the router in a reasonably central, open location for better coverage.
- Avoid damaging, sharply bending, or pulling network cables by their connectors.
- Label cables at both ends in a lab or office to make troubleshooting easier.

Conclusion

A reliable home network depends on several functions working together. The modem or ONT provides the ISP connection, the router directs traffic between networks, the switch adds wired connectivity, and the access point allows wireless devices to join the LAN. Ethernet, coaxial, and fiber-optic cables serve different roles depending on the connection type and network requirements. Knowing the difference between a router and a switch is an important foundation for learning networking and cybersecurity.

References / Further Reading

1. Cisco — What is Computer Networking? https://www.cisco.com/site/us/en/learn/topics/networking/what-is-computer-networking.html

2. Cisco — What is Ethernet? https://www.cisco.com/site/us/en/learn/topics/networking/what-is-ethernet.html

3. North Carolina State University — Wired and Wireless Connections (cable types and images). https://e115.engr.ncsu.edu/networking/network-media/

4. Cisco — What is an Ethernet Switch? https://team-development.cisco.com/site/us/en/learn/topics/networking/what-is-an-ethernet-switch.html

5. Cisco — What is a Router? https://team-development.cisco.com/site/us/en/learn/topics/small-business/what-is-a-router.html

End of Assignment